Data-Driven COVID-19 Detection & Trend Prediction Using Machine Learning

Team Members	Domain
Anusha Dhale - 24B2209 Shravanee Kulkarni - 24B2206 Sparsh Gupta - 24B2103 Sparsh Tulsyan - 24B2185	Biomedical Imaging + Epidemiological Time-Series Forecasting

1. The Scientific & Engineering Problem (Biomedical Imaging + Epidemiological Time-Series Forecasting)

1.1 Clinical Detection - Chest X-Ray Classification

Multi-class classification to distinguish COVID-19 from other pulmonary conditions where visual similarities make manual diagnosis difficult. Early diagnosis is critical, but PCR testing is slow and resource-intensive

1.2 Epidemiological Forecasting - Case Trajectory Prediction

Captures local temporal dynamics and epidemic trends. Classical ARIMA models fail to capture non-stationary effects like interventions and variants.

2. Base Work & Reference

- **CXR Detection:** Building on *varshh/covid-19-detection (Kaggle)* by replacing a custom CNN with a fine-tuned **EfficientNet-B4** and adding a **Grad-CAM** explainability layer.
- **Forecasting:** Extending *midouazerty/covid-19-cases-prediction (Kaggle)* using a **Hybrid Transformer-LSTM** ensemble with multi-country intervention covariates.

3. Dataset Sources

- **COVID-19 Radiography Database:** 21,165 CXR images for image classification.
- **OWID / JHU CSSE:** Daily cases, deaths, and vaccinations across 200+ countries.
- **Google Health Open Data:** Mobility, weather, and demographic covariates for exogenous features.

4. Proposed Methodology

4.1 Module 1 - (Vision) COVID-19 X-Ray Classifier

Uses Transfer Learning with EfficientNet-B4. Pre-processing includes CLAHE normslization and data augmentation. The training strategy involves freezing the base model for 5 epochs before fine-tuning.

4.2 Module 2 - Epidemic Forecasting (Transformer-LSTM Hybrid)

Model integrates endogenous features like daily cases and deaths with exogenous features, including mobility data and government stringency indices, to generate 7, 14, and 30-day horizons with uncertainty intervals.

5. Success Criteria

- **Vision Validation:** Targets **93% accuracy and 95% recall**; uses an **80/10/10 patient-level split** and Grad-CAM for clinical verification.
- **Forecasting Validation:** Targets a **MAPE < 7%** for 7-day forecasts; uses **walk-forward monthly retraining** and back testing across major COVID waves.